

Because open ocean aquaculture is at an early development stage in New Zealand, carefully designed monitoring programmes will be required to assess the effectiveness of mitigation measures. The Guidelines therefore consider the range of monitoring options that may be appropriate in open ocean environments.

Effective monitoring will inform structural or operational improvements that deliver better environmental outcomes over time. The flexibility to evolve both monitoring and farming practices will be essential given the lack of New Zealand and international experience with open ocean aquaculture (OOA).

Guideline development process:

- Literature reviews were commissioned on the effects of open ocean aquaculture on **seabirds and marine mammals**. These provided a starting point for developing best practice guidelines. Following this, the Aquatic Environment Working Group and the Aquaculture Working Group (chaired by Fisheries New Zealand) provided peer review of the first and second drafts of this document.
- Fisheries New Zealand commissioned a report focused mainly on benthic effects of open ocean aquaculture overseas. This included several case studies from Norway regarding water quality effects. A technical working group was formed to discuss issues and develop guidelines for **benthic and water column effects** given the range of options available for monitoring and management of these effects.
- **Guidelines will be reviewed as needed** based on growing knowledge, so they can be improved. Considering this is the early days of open ocean aquaculture in New Zealand (and elsewhere), it is envisaged the initial review cycles will be relatively short with a backstop of 5 years from the development of the farms, or as required (e.g., if monitoring data and results are significantly different from anticipated).
- **Further research** is required into the effects of open ocean aquaculture upon the marine environment. In addition to known information gaps, research needs will be identified and documented as part of the guideline development and review process. These can then input to any ongoing or future Fisheries New Zealand research planning or prioritisation processes such as the proposed Aquaculture Innovation Plan and the proposed Aquaculture Strategy and Investment Roadmap, currently under development.

Factors to consider when reading and applying the Guidelines:

- Fisheries New Zealand is leading (with input from other government agencies) a process of considering options **for a future regulatory framework for open ocean aquaculture**. This future **open ocean aquaculture regulatory framework will take into consideration** concurrent regulatory change including the Biosecurity Act review, marine protected areas reform, and particularly the review of the Resource Management Act with a new Natural and Built Environment Act and Strategic Planning Act.
- While this regulatory framework may lead to recommended changes for any new farms proposed, **these guidelines are expected to remain relevant under any future regulatory framework**. The guideline review process can be used to manage required adaptations. Although we describe these guidelines within a Resource Management Act context, they are technical guidelines that will remain relevant under any regulatory framework.
- For the purpose of developing guidelines, **open ocean aquaculture is defined as “aquaculture outside of semi-enclosed bays and harbours or other sheltered locations around mainland New Zealand and larger offshore islands”**. This definition